

2ÈME ANNÉE CYCLE D'INGÉNIEUR - SPÉCIALITÉ INFORMATIQUE.

MATIÈRE :
BASES DE DONNÉES AVANCÉES

Compte rendu TP1

Réalisé par :
BENMESSAOUD Abdellah

Encadrés par :
Mr. Samir Youcef

Table des matières

1 Exercice n°1	2
1.1 ajout de la table section	2
1.2 Diagramme entité-association	2
1.3 Insertion de nouveau cours	2
2 Exercice N°2	3

1 Exercice n°1

1.1 ajout de la table section

```
create table section
(
    course_id      varchar(8),
    sec_id         varchar(8),
    semester       varchar(6),
    check (semester in ('Fall', 'Winter', 'Spring', 'Summer')),
    year           numeric(4,0),
    building        varchar(15),
    room_number     varchar(7),
    time_slot_id    varchar(4),
    primary key (course_id, sec_id, semester, year),
    foreign key (course_id) references course,
    foreign key (building, room_number) references classroom
);
```

FIGURE 1

1.2 Diagramme entité-association

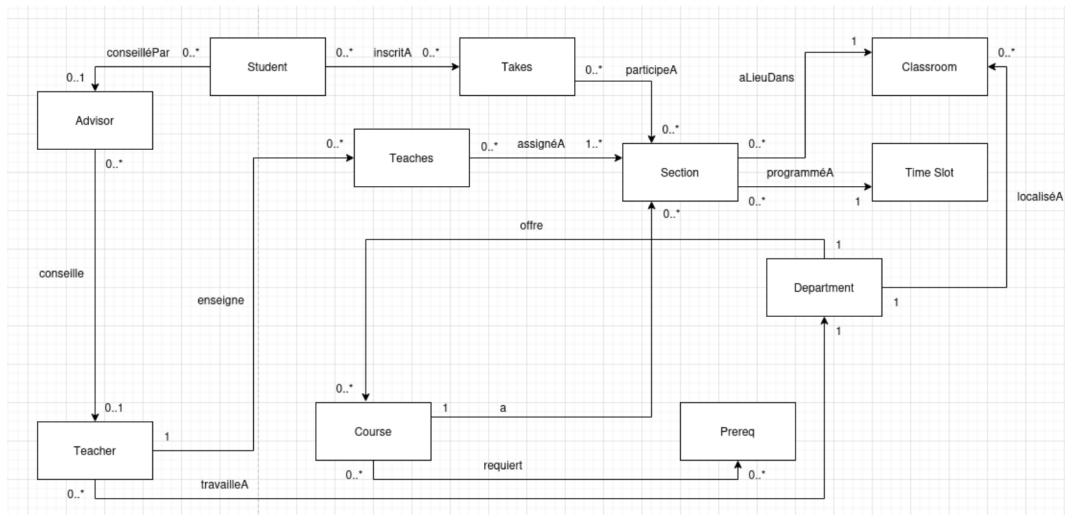


FIGURE 2

1.3 Insertion de nouveau cours

```
insert into course values ('BI0-101', 'Intro. to Biology', 'Biology', '4')
```

FIGURE 3

2 Exercice N°2

- 1) DESCRIBE section;
- 2) SELECT * FROM course;
- 3) SELECT title, dept_name FROM course;
- 4) SELECT dept_name, budget FROM department;
- 5) SELECT name, dept_name FROM teacher;
- 6) SELECT name FROM teacher WHERE salary > 65000;
- 7) SELECT name FROM teacher WHERE salary BETWEEN 55000 AND 85000;
- 8) SELECT DISTINCT dept_name FROM teacher;
- 9) SELECT name FROM teacher
WHERE dept_name = 'Comp. Sci.' AND salary > 65000;
- 10) SELECT * FROM section
WHERE semester = 'Spring' AND year = 2010;
- 11) SELECT title FROM course
WHERE dept_name = 'Comp. Sci.' AND credits > 3;
- 12) SELECT teacher.name, teacher.dept_name, department.building
FROM teacher
JOIN department ON teacher.dept_name = department.dept_name;
- 13) SELECT DISTINCT student.name
FROM student
JOIN takes ON student.ID = takes.ID
JOIN section ON takes.course_id = section.course_id
JOIN course ON section.course_id = course.course_id
WHERE course.dept_name = 'Comp. Sci.';
- 14) SELECT DISTINCT student.name
FROM student
JOIN takes ON student.ID = takes.ID
JOIN teaches ON takes.course_id = teaches.course_id
AND takes.sec_id = teaches.sec_id
JOIN teacher ON teaches.ID = teacher.ID
WHERE teacher.name = 'Einstein';